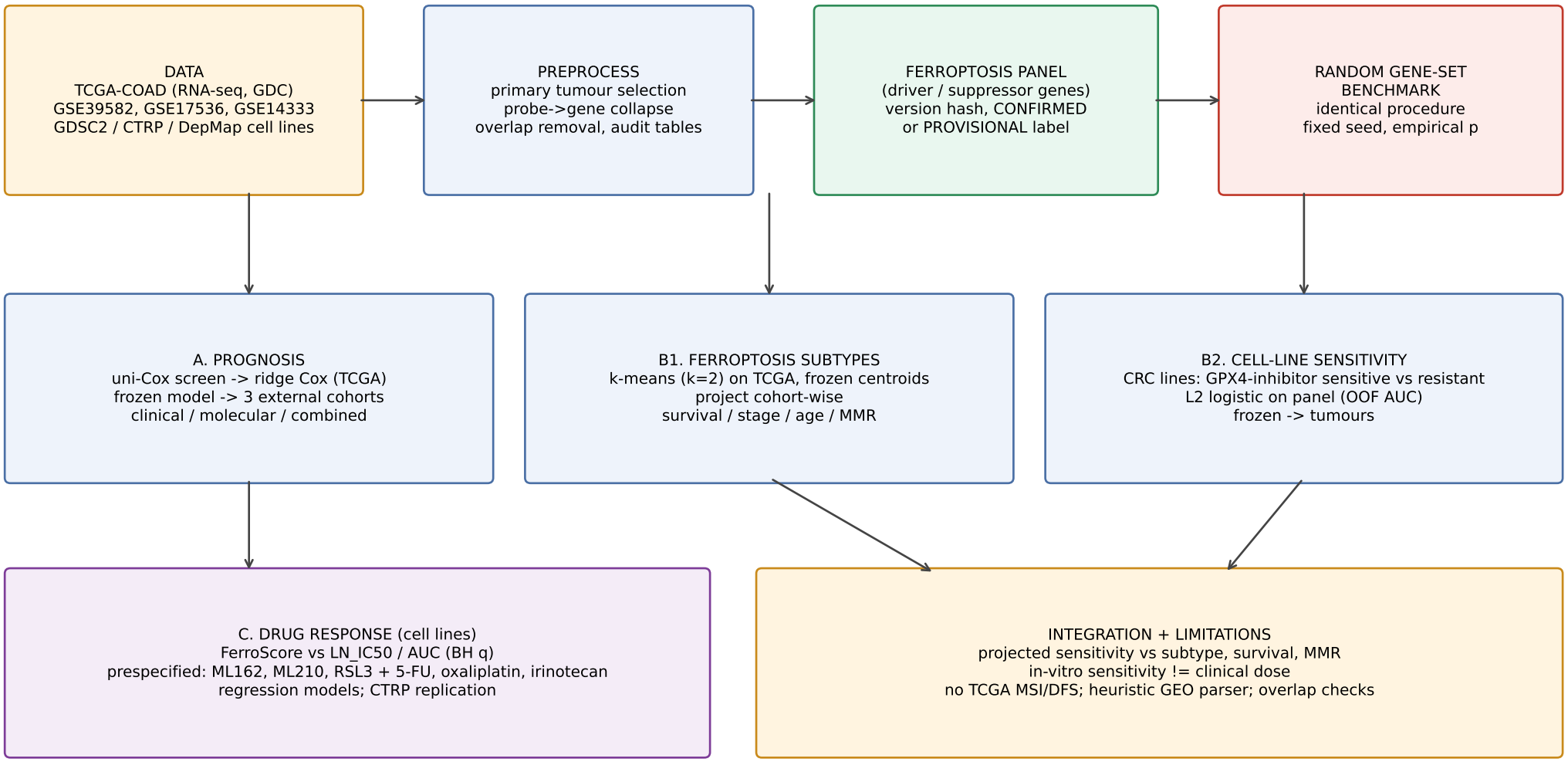
Ferroptosis classification, prognosis and cell-line sensitivity pipeline (colorectal cancer)
DATA
TCGA-COAD (RNA-seq, GDC)
GSE39582, GSE17536, GSE14333
GDSC2 / CTRP / DepMap cell lines
PREPROCESS
primary tumour selection
probe->gene collapse
overlap removal, audit tables
FERROPTOSIS PANEL
(driver / suppressor genes)
version hash, CONFIRMED
or PROVISIONAL label
RANDOM GENE-SET
BENCHMARK
identical procedure
fixed seed, empirical p
A. PROGNOSIS
uni-Cox screen -> ridge Cox (TCGA)
frozen model -> 3 external cohorts
clinical / molecular / combined
B1. FERROPTOSIS SUBTYPES
k-means (k=2) on TCGA, frozen centroids
project cohort-wise
survival / stage / age / MMR
B2. CELL-LINE SENSITIVITY
CRC lines: GPX4-inhibitor sensitive vs resistant
L2 logistic on panel (OOF AUC)
frozen -> tumours
C. DRUG RESPONSE (cell lines)
FerroScore vs LN_IC50 / AUC (BH q)
prespecified: ML162, ML210, RSL3 + 5-FU, oxaliplatin, irinotecan
regression models; CTRP replication
INTEGRATION + LIMITATIONS
projected sensitivity vs subtype, survival, MMR
in-vitro sensitivity != clinical dose
no TCGA MSI/DFS; heuristic GEO parser; overlap checks